ASI: Building Distinct Pillars

What we are changing in the pillars, how we will optimise them, and the statistics behind it

17 October 2026. An explainer for the owner, who will rule on the indicator list and work through the optimisation. Every number comes from command output on the ASI panel (edition 2, step 1, reference year 2024) and the WDI and WGI datasets in Athena.

What	Count
Candidate indicators screened from WDI and WGI	98 (95 with data in 2020–24)
…reported by at least 33 of the 54 states	77
…closer to their own pillar than to any other	45
Candidates from other sources, listed for ingest	21
Mean correlation between pillars today (|ρ|)	0.515
…with each indicator counted in one pillar only	0.365

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1. In one page

- **The goal.** Keep the seven pillars, let each become broader (semi-comprehensive), and make each measure something the others do not. Overlapping pillars count the same thing twice and hide the real weights.
- **Where we start.** The pillars correlate at |ρ| = 0.515 on average; the closest pair, A and G, at 0.81.
- **Most of that overlap is structural, not statistical.** Five WGI indicators are counted in two pillars each. Counting each indicator once lowers the mean to 0.365, and A–E falls from .79 to .25. Water and sanitation split across Health and Infrastructure keep D–G at .81.
- **The third variable is development.** Holding GDP per capita constant lowers the mean to 0.354; with both changes it is 0.239.
- **The right target is not zero.** You ruled that the pillars measure stability (P48), so they should share some common ground. The target is *discriminant validity*: no pillar a copy of another (HTMT below .85), and every indicator closer to its own pillar than to any other.
- **How.** Choose and place indicators by what they mean first, then test them with the ADA toolkit: correlations with their samples and intervals, partial correlations, reliability and attenuation, and factor analysis. Validate on other years so we do not fit to noise.
- **The list.** 12 keep, 15 add, 37 test, 3 move, 3 replace, 18 descriptive, 10 drop, and 21 from new sources (section 7).
- **Your part.** Rule on the list, then work through the Test items with me in optimisation sessions; each change shows its effect before you approve it.

2. The goal, and a top-down evaluation

2.1 What independent pillars buy us

A composite index adds pillars together. If two pillars move in step, adding them counts one underlying thing twice: the index leans towards it whatever weights we write down. Distinct pillars mean three things:

- **Honest weights.** Equal weights are only equal if the pillars are not copies of each other. Today the six WGI indicators, counted in A and again in C, E and G, carry 28.9% of the composite (methodology review).
- **Readable profiles.** A country can be strong on governance and weak on health; that only shows if the pillars measure different things.
- **Information.** A pillar that repeats another adds nothing a reader could not already see.

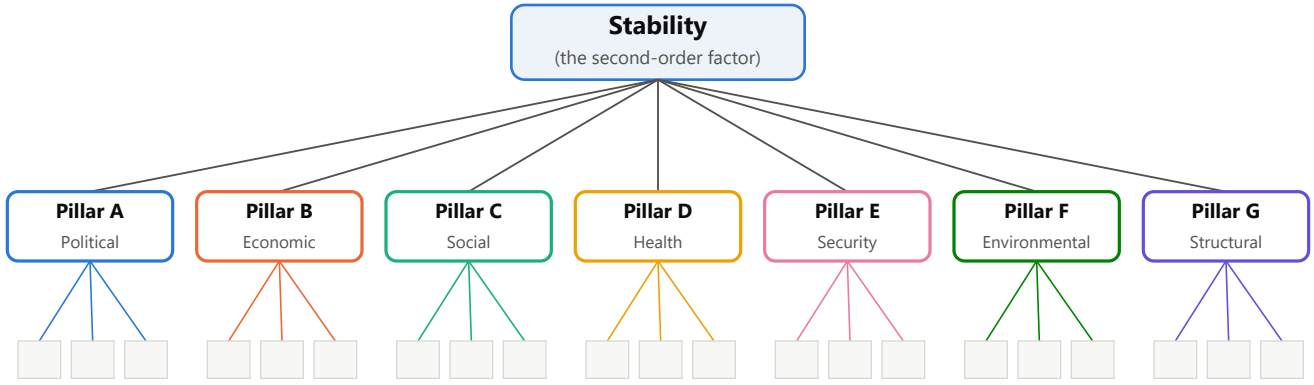
2.2 Why zero correlation is neither possible nor the target

Two facts set the ceiling on independence:

- **Development pulls everything together.** Richer states tend to be better governed, healthier, better connected. That is a real feature of the world, not a defect of the index. Of the candidates screened, 18 correlate at $|\rho| \geq .70$ with GDP per capita.
- **You ruled the pillars measure stability (P48).** If each pillar reflects stability, they must correlate somewhat: that shared part is stability itself. Forcing them to zero would remove the very thing measured.

The model that reconciles both is a **second-order factor model** (figure 1): stability at the top, seven pillars beneath it, each pillar measured by its own indicators. Pillars may correlate through stability, but each must carry something of its own. In practice the target becomes:

> **Target.** (1) every indicator correlates more with its own pillar than with any other; (2) no pair of pillars has an HTMT ratio above .85 (section 5.5); (3) the mean $|\rho|$ between pillars is reported, as published and with GDP held constant, and pushed down where meaning allows.



Indicators: each belongs to one pillar and should correlate with its own pillar more than with any other (convergent and discriminant validity).
Pillars: may correlate through the stability they all reflect, but none should be a copy of another (HTMT below .85).
The common pull of development (GDP) is checked separately, as a third variable.

Figure 1. The second-order model: distinct pillars that all reflect stability.

2.3 What could go wrong, and the guards

Risk	What it looks like	Guard
Overfitting (data dredging)	Choosing indicators because they lower a correlation this year, not because they measure the pillar	Meaning first: an indicator enters only with a written reason. Rules agreed before the numbers are seen (step 0). Every change re-tested on other years (2015, 2019).

Risk	What it looks like	Guard
Small sample	54 states: a correlation's 95% interval is often half a unit wide (median .57 at 2023)	Report n and intervals for every figure; decide only on differences larger than their uncertainty; use the bootstrap.
Losing content	Dropping everything development-related would make the index blind to real differences	Two views: the index as built, and a development-adjusted view; never strip GDP out silently.
Goodhart's law	Indicators governments can move without changing reality	Prefer outcomes and hard counts (deaths, completion) over inputs and ratings; the advisors' Doing Business warning stands.
Missing data not at random	The least stable states report least	Coverage floor of 33; no regional-mean fill in scoring (approved plan 4.5); a reliability index beside every score.
Gaps read as zeros, or zeros read as gaps	WDI leaves conflict series blank where there was no conflict	Take conflict data from UCDP and IDMC, where no record means zero.

2.4 Verdict

The plan is feasible, and the largest gains are structural rather than clever: one home per indicator, water and sanitation in one pillar, conflict measured directly instead of through WGI perceptions, and a few indicators whose direction the African data contradicts. Statistics then confirm and fine-tune. Orthogonalising the pillars mathematically (rotating them until they are uncorrelated) is possible but not recommended: the rotated pillars stop meaning governance, health and so on (section 6.3).

3. What the data shows today

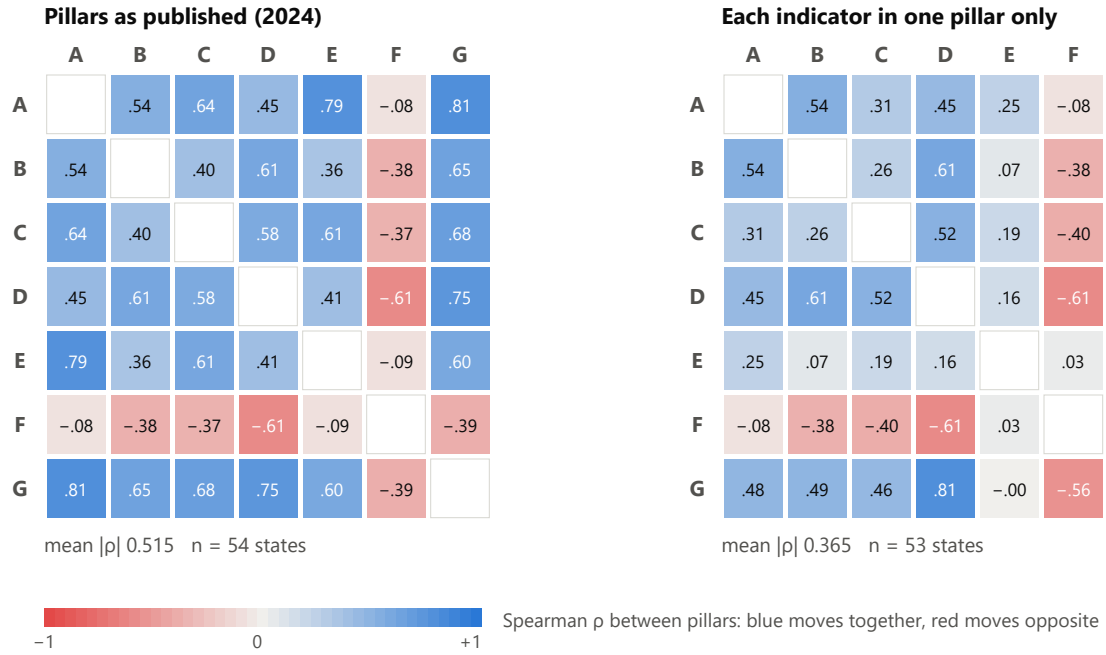


Figure 2. Correlations between pillars in 2024. Left: as published. Right: each WGI indicator kept in its first pillar only, so none is counted twice.

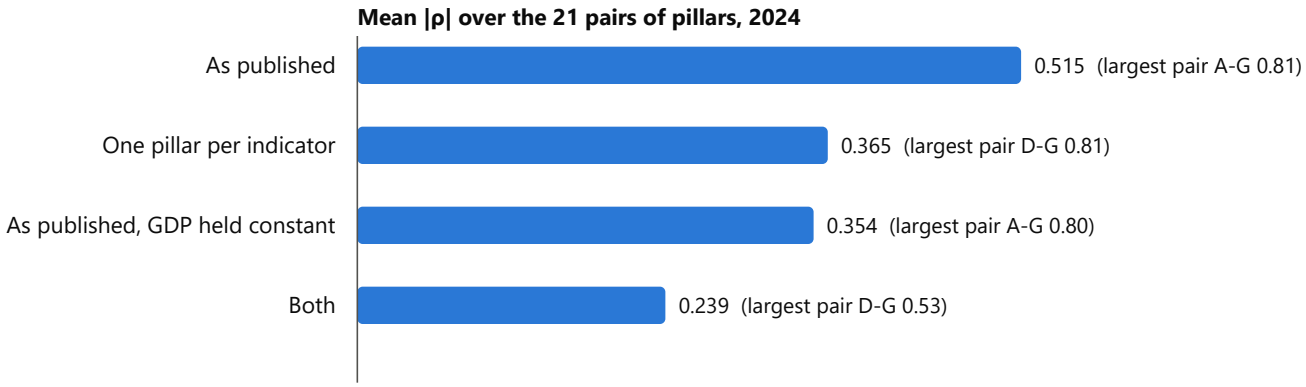


Figure 3. Mean absolute correlation between pillars under four scenarios.

- **Cross-listing is the biggest single source of overlap.** Voice and accountability sits in A and C; political stability and rule of law in A and E; government effectiveness and control of corruption in A and G. Counting each once lowers the mean from .515 to .365.
- **Water, sanitation and hygiene are split across D and G.** After de-duplication, D–G is the closest pair (.81). Putting all of them in one pillar is the next structural fix.
- **GDP is the third variable.** Partialling it out lowers the mean to .354 as published, and .239 with de-duplication; the largest remaining pair falls to .53.
- **Some directions are wrong for Africa.** Measured unemployment is *higher* in richer states (aligned ρ with Pillar B −.11; youth −.24), because in poor agrarian economies few can afford to be unemployed. Female labour participation and child wasting show the same reversal. They become descriptive or are dropped.
- **Total GDP measures size.** It is barely related to Pillar B (ρ = .03) and most related to E (.24). As a scored indicator (your ruling) it adds information nothing else carries, which helps distinctness; read it as the size of the economy a government taxes and spends from.

- **Women in parliament correlates with nothing** (ρ = .01 with A). Under the Interpeace definition it is part of inclusion; statistically it is new information rather than a copy.

4. How other composite indices build their pillars

Index	How the pillars are formed	What we take from it
Human Development Index (UNDP)	Three dimensions, one or two indicators each, geometric mean	Few, strong, outcome indicators per dimension
Social Progress Index	Three dimensions, 12 components, outcome indicators only; deliberately excludes economic indicators so social progress can be compared against GDP	Outcomes not inputs (we drop spending ratios); the idea of a development-adjusted view
Global Innovation Index, audited by the EU JRC	Seven pillars; the JRC audit checks statistical coherence with correlation analysis, principal components and Cronbach's α, and adds confidence intervals to ranks	Each indicator tested against its own pillar and the others; intervals on ranks
INFORM Risk (EC JRC, UN OCHA)	Three dimensions; geometric means; fixed goalposts; a separate Lack of Reliability Index	Reliability beside every score; 5-year recency rule
OECD States of Fragility	Six dimensions; principal components within each dimension, then clustering into groups rather than ranks	Groups and tiers instead of point ranks
Ibrahim Index of African Governance	Four categories; indicators must cover at least 33 African countries; outliers capped beyond 3 trimmed SDs	Our coverage floor and outlier rule
Fragile States Index	Twelve indicators in four groups; data streams triangulated	Triangulation: more than one source family per pillar
Global Peace Index	23 indicators; weights from an expert panel; internal 60%, external 40%	Weights can be a judgement, stated openly
ND-GAIN	Vulnerability (exposure, sensitivity, capacity) and readiness	A structure for Pillar F: exposure versus capacity
WGI (World Bank)	Unobserved components model with a standard error for every estimate	Uncertainty published with every number

Common practice, in short: define the dimensions by theory; prefer outcome indicators; check each dimension's internal coherence (α, item-rest correlations); check the structure with correlations and principal components; publish uncertainty. None of them forces dimensions to be uncorrelated; the audits ask that each indicator belong more to its own dimension than to others, which is our target.

5. The concepts, explained

Each concept comes from the ADA course or extends it. For each: what it is, an ASI example, and how we use it.

5.1 Correlation, r^2, and Spearman's ρ

Correlation measures how two variables move together, from −1 to +1. r^2 is the share of one variable's variation the other accounts for: r = .70 means 49% shared. Spearman's ρ correlates ranks instead of values, so it is not thrown by skew or outliers; we use it to screen.

In the ASI: Pillars A and G, ρ = .81: about two thirds of their rank variation is shared.

5.2 Sample size and the interval around r

A correlation from 16 countries is far less certain than one from 54. Fisher's transformation gives r an interval: convert r to z′, add ±1.96 standard errors, convert back.

```
z′ = artanh(r)  SE = 1 / √(n − 3)  r interval = tanh(z′ ± 1.96·SE)
```

In the ASI: At 2023 the median interval between two indicators was .57 wide; r = .50 is not significant at n = 14 but is at n = 20.

5.3 Third variables: partial and semi-partial correlation

A partial correlation removes a third variable Z from both X and Y. If X and Y correlate only because both follow Z, the partial falls to zero (spurious); if it rises, Z was hiding the link (suppression).

```
r(xy·z) = (r_xy − r_xz·r_yz) / √((1 − r_xz²)(1 − r_yz²))
```

In the ASI: Electricity access and handwashing: ρ = .68, but .04 with GDP per capita held constant.

5.4 Reliability and attenuation

Cronbach's α measures how consistently a pillar's indicators move together. Measurement error lowers correlations; correcting for it estimates the correlation between the true scores. A corrected value near or above 1 means two pillars measure the same thing.

```
r_true = r / √(α_x · α_y)
```

In the ASI: Corrected for reliability, A–E reached 1.22 in 2023: one measurement taken twice.

5.5 Convergent and discriminant validity

Convergent: an indicator correlates with the rest of its own pillar (item-rest ρ ≥ .30). Discriminant: it correlates less with every other pillar. At pillar level, the heterotrait-monotrait ratio (HTMT; Henseler, Ringle and Sarstedt, 2015) compares correlations between pillars' indicators with correlations within them; below .85 (strict) or .90 (lenient) the pillars are distinct. The older Fornell–Larcker test is weaker.

```
margin = |ρ own pillar| − max |ρ other pillar|  HTMT(i,j) = mean ρ between / √(mean ρ within_i · within_j)
```

In the ASI: Of 95 candidates, 45 are closer to their own pillar than to any other (figure 4).

5.6 Factor analysis

Factor analysis asks how many underlying dimensions explain the correlations. KMO and Bartlett's test say whether the data are factorable; parallel analysis keeps only factors stronger than random data would give; rotation (varimax, or oblique when factors may correlate) makes loadings readable. Confirmatory factor analysis then tests a stated structure, such as figure 1.

In the ASI: At pillar level in 2023: KMO .784, one factor as scored, two with GDP held constant. At indicator level the panel has only 31 complete states for 32 indicators, too few; so we work pillar by pillar and pool years with care.

5.7 Redundancy

Two indicators correlating at $|\rho| \geq .90$ are near-duplicates: keeping both doubles their weight.

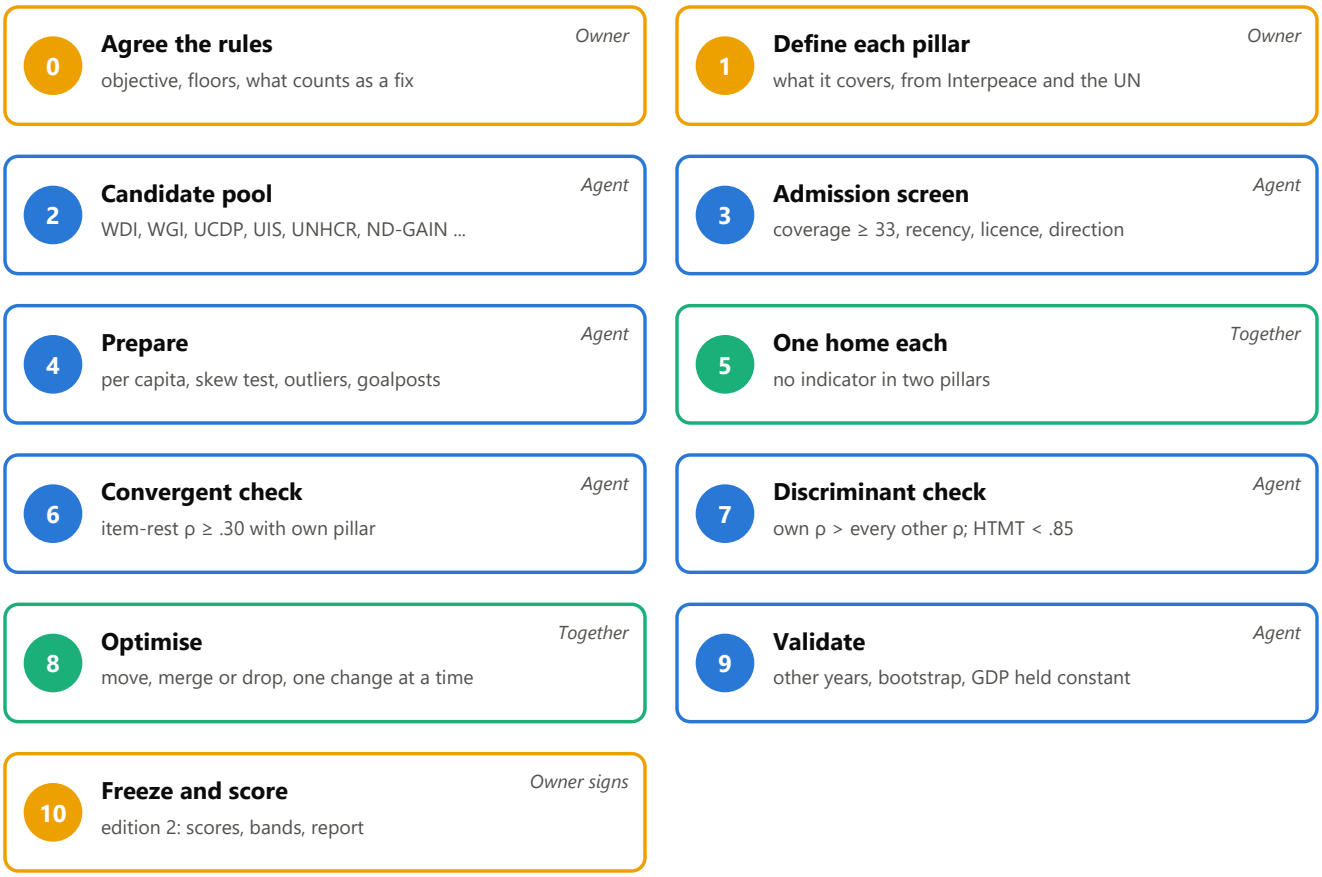
In the ASI: Rule of law and control of corruption; infant and under-5 mortality.

5.8 Guarding against overfitting

Any choice made by looking at this year's correlations may fit chance. We re-check every structure on other years and on bootstrap samples (countries resampled with replacement) and keep only what holds.

In the ASI: A change that lowers HTMT in 2024 but not in 2015 or 2019 is rejected.

6. The optimisation procedure



Read left to right, then down. Steps 6 to 8 repeat until the checks hold; step 9 can send the structure back to step 8.

Figure 5. The procedure, step by step, and who leads each step.

6.1 The rules, stated before any numbers are looked at

- **Objective:** lower the largest HTMT between pillars, then the mean |ρ|, in the index as built.
- **Constraints:** each pillar keeps at least 4 scored indicators from at least 2 source families; every indicator covers 33 or more states and has a written reason tied to its pillar's definition; a pillar's definition is never narrowed to win a correlation.
- **Moves allowed:** place an indicator in the one pillar it fits; merge near-duplicates; replace a weak source with a direct one; make an indicator descriptive. Each move is shown with its before-and-after statistics and approved by you.
- **Acceptance:** a change stays only if it also holds in 2015 and 2019 and in 1,000 bootstrap samples.

6.2 What happens at each step

Step	Statistic (ADA concept)	Passes when
3 Admission	n with every figure (5.2); skew z-test	n ≥ 33 in 2020–24; median year ≥ 2018; licence read; direction confirmed in the data
4 Prepare	skew/SE(skew); |z| > 3 outliers	skewed counts logged; outliers capped before goalposts
5 One home	—	every indicator in exactly one pillar
6 Convergent	item-rest ρ with interval (5.1, 5.2)	ρ ≥ .30, interval clear of zero
7 Discriminant	margin; HTMT; corrected r (5.4, 5.5)	margin > 0; HTMT < .85
7b Third variable	partial ρ with log GDP (5.3)	the pillar keeps its meaning with GDP held constant

Step	Statistic (ADA concept)	Passes when
8 Structure	parallel analysis, CFA (5.6)	seven distinct factors, or a stated reason why not
9 Validate	bootstrap; two-correlation Z test across years	the pattern holds; no difference beyond chance

6.3 Other ways to decorrelate, and why selection comes first

Method	What it does	Verdict
Selection and placement	Choose and place indicators so pillars measure different things	**Main method.** Keeps meaning; explainable
Development-adjusted view	Regress each pillar on log GDP per capita and score the residual: stability beyond what income predicts	**Second lens**, published beside the main index
Orthogonal rotation or PCA	Mathematically uncorrelated components	Not recommended: components stop meaning governance, health ...
Weights	Down-weight overlapping pillars	Treats the symptom; used only as a sensitivity check
Trimming data	Removing outlying states or years	Only for errors or extreme outliers, capped (winsorised), never deleted to improve a statistic

7. The proposed indicator list

Each candidate carries a first-position status for you to rule on. **Keep**: scored now and staying. **Add**: new, passes the screens. **Test**: decided in the optimisation sessions. **Move**: belongs in another pillar. **Replace**: same idea, better source. **Descriptive**: shown, not scored. **Drop**: removed. Columns: n is the number of states with a real value in 2020–24; own ρ is the Spearman correlation with its current pillar (inflated for indicators already scored in it); closest other is the other pillar it correlates with most; GDP ρ is its correlation with log GDP per capita. Signs are aligned so that positive means more stable.

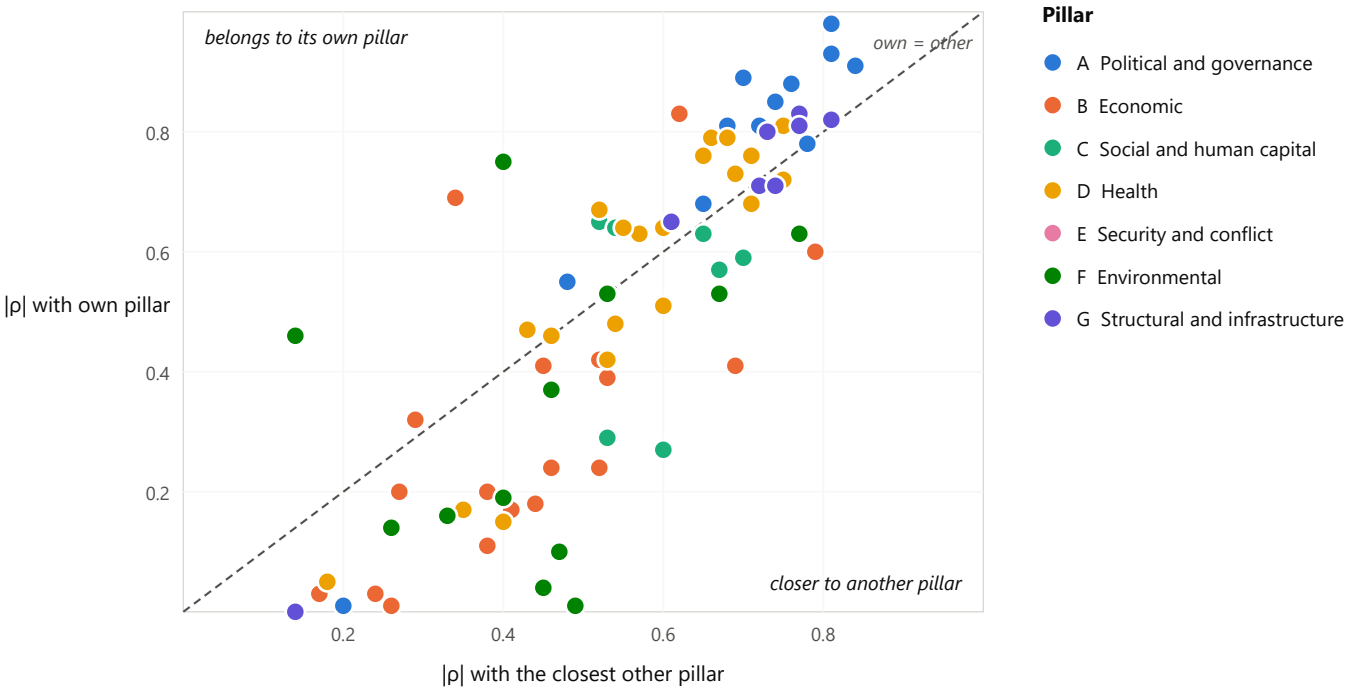


Figure 4. Every candidate with 33 or more states: correlation with its own pillar against its closest other pillar. Above the line it belongs where it is.

7.1 Pillar A: Political and governance

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Government effectiveness (WGI) `GOV_WGI_GE.EST` (scored today)	**Keep**	54	+0.91	G 0.84	+0.54	Stays in A only; no longer cross-listed in G
Political stability, no violence (WGI) `GOV_WGI_PV.EST` (scored today)	**Keep**	54	+0.78	E 0.78	+0.43	Stays in A only; no longer cross-listed in E
Rule of law (WGI) `GOV_WGI_RL.EST` (scored today)	**Keep**	54	+0.98	G 0.81	+0.43	Justice for all (Interpeace); stays in A only, no longer in E
Voice and accountability (WGI) `GOV_WGI_VA.EST` (scored today)	**Keep**	54	+0.89	E 0.70	+0.25	Anchor of political inclusion and accountability
Statistical performance `IQ.SPI.OVRL`	**Add**	53	+0.55	G 0.48	+0.16	State capacity to know its own population; distinct (GDP rho .16)
Women in parliament `SG.GEN.PARL.ZS`	**Add**	52	+0.01	F 0.20	0.00	Inclusion of women (Interpeace). Barely correlated with anything: new information
CPIA property rights and rule-based governance `IQ.CPA.PROP.XQ`	**Test**	41	+0.85	E 0.74	+0.52	Adds a second source family to A; covers 41 IDA-eligible states only

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
CPIA public sector management IQ.CPA.PUBS.XQ	**Test**	41	+0.81	G 0.68	+0.50	Near-duplicate of CPIA transparency: keep at most one
CPIA transparency and accountability IQ.CPA.TRAN.XQ	**Test**	41	+0.81	G 0.72	+0.53	Expert-assessed, not perception-based; near-duplicate of the CPIA public-sector rating
Control of corruption (WGI) GOV_WGI_CC.EST (scored today)	**Test**	54	+0.93	G 0.81	+0.43	Near-duplicate of rule of law (rho >= .90): keep one
Regulatory quality (WGI) GOV_WGI_RQ.EST (scored today)	**Test**	54	+0.88	G 0.76	+0.43	Near-duplicate of government effectiveness (rho >= .90): keep one
Tax revenue (% GDP) GC.TAX.TOTL.GD.ZS	**Test**	33	+0.68	G 0.65	+0.28	The social contract; exactly at the 33-state floor
Rule of Law Index World Justice Project	**Ingest, terms first**	—	—	—	—	Justice as experienced
Share of the population politically excluded Ethnic Power Relations, ETH Zurich	**Ingest, terms first**	—	—	—	—	Representation of all groups
Trust in president, parliament, police Afrobarometer	**Context or validation**	—	—	—	—	The state-society relationship; ~39 states every 2-3 years
V-Dem participatory and egalitarian indices V-Dem Institute	**Ingest, terms first**	—	—	—	—	Inclusion and equal access to power (Interpeace)

7.2 Pillar B: Economic

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
GDP per capita, PPP (log) NY.GDP.PCAP.PP.CD (scored today)	**Keep**	52	+0.60	G 0.79	+1.00	Owner's ruling P50; the development axis itself, watched as the third variable
Inflation (5-year average) FP.CPI.TOTL.ZG (scored today)	**Keep**	50	+0.69	G 0.34	+0.31	Most distinct economic indicator (margin .36)
Private credit (% GDP) FS.AST.PRVT.GD.ZS (scored today)	**Keep**	48	+0.83	G 0.62	+0.38	Distinct (margin .21)
GDP, total (log) NY.GDP.MKTP.CD	**Add**	52	+0.03	E 0.24	+0.24	Owner's ruling: scored. Measures the economy's size; adds information nothing else carries
Account ownership (adults) FX.OWN.TOTL.ZS	**Move to G**	41	+0.24	A 0.52	+0.31	Financial inclusion is an access-to-services measure; closer to A than B
Debt service (% GNI) DT.TDS.DECT.GN.ZS	**Test**	48	-0.41	G 0.45	-0.51	As external debt; the fiscal squeeze on what a government can spend
External debt (% GNI) DT.DOD.DECT.GN.ZS	**Test**	48	-0.18	A 0.44	-0.11	Fiscal risk in theory, but richer states borrow more: check with GDP held constant
FDI inflows (% GDP) BX.KLT.DINV.WD.GD.ZS	**Test**	52	+0.17	E 0.41	0.00	Weak link to B; volatile year to year
Fuel exports (% merchandise) TX.VAL.FUEL.ZS.UN	**Test**	44	+0.20	E 0.38	-0.14	Commodity dependence (structural cause, fair framing); weakly related to B
GDP growth (3-year average) NY.GDP.MKTP.KD.ZG (scored today)	**Test**	52	+0.20	F 0.27	-0.24	Weak link to its own pillar (rho .20)
Natural resource rents (% GDP) NY.GDP.TOTL.RT.ZS	**Test**	52	+0.42	A 0.52	+0.25	Resource dependence; closer to A
Ore and metal exports (% merchandise) TX.VAL.MMTL.ZS.UN	**Test**	44	+0.03	E 0.17	-0.07	Commodity dependence; weakly related to B

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Trade (% GDP) NE.TRD.GNFS.ZS	**Test**	49	+0.32	G 0.29	+0.32	Connectivity of goods (Interpeace); weak links everywhere
Vulnerable employment SL.EMP.VULN.ZS	**Test**	53	+0.41	G 0.69	+0.74	Informality is the real African labour measure; closer to G and development-driven
Aid received (% GNI) DT.ODA.ODAT.GN.ZS	**Descriptive**	51	+0.39	D 0.53	+0.85	Measures poverty more than stability (GDP rho .85)
Gini index SI.POV.GINI (scored today)	**Descriptive**	27	+0.41	F 0.39	+0.16	27 states in 2020-24, below the floor
Poverty at $3.00 a day SI.POV.DDAY	**Descriptive**	27	+0.66	G 0.67	+0.78	27 states in 2020-24
Unemployment SL.UEM.TOTL.ZS	**Descriptive**	53	-0.11	G 0.38	-0.52	Direction contradicted in Africa: measured unemployment is higher in richer states (informality)
Unemployment, advanced education SL.UEM.ADVN.ZS	**Descriptive**	37	-0.01	E 0.26	-0.10	The advisors' 'educated but jobless'; direction unclear, 37 states
Youth unemployment SL.UEM.1524.ZS	**Descriptive**	53	-0.24	D 0.46	-0.60	Same direction problem as total unemployment; shown for youth (Interpeace)
Food price inflation FAOSTAT	**Ingest, terms first**	—	—	—	—	Food price shocks (Dr. Smyth's list)
Government debt and revenue IMF World Economic Outlook	**Ingest, terms first**	—	—	—	—	Fiscal space (WDI's debt series covers 13 states)

7.3 Pillar C: Social and human capital

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Youth literacy SE.ADT.1524.LT.ZS (scored today)	**Keep**	34	+0.65	G 0.52	+0.54	Near-duplicate of adult literacy: keep youth, drop adult
Adolescent fertility SP.ADO.TFRT	**Add**	54	+0.57	D 0.67	+0.52	Girls' and young women's opportunity (Interpeace youth and women); full coverage
Primary completion SE.PRM.CMPT.ZS	**Add**	40	+0.63	G 0.65	+0.69	Outcome replacing enrolment; ties with G, watch
Lower-secondary completion SE.SEC.CMPT.LO.ZS	**Test**	38	+0.59	G 0.70	+0.78	Closer to G and development-driven; keep one completion rate
Birth registration SP.REG.BRTH.ZS	**Descriptive**	6	+0.77	F 0.89	+0.26	6 states
Female homicide rate VC.IHR.PSRC.FE.P5 (scored today)	**Descriptive**	16	+0.59	D 0.44	+0.05	16 states; belongs with violence in E if sources improve
Gender parity, primary SE.ENR.PRIM.FM.ZS (scored today)	**Descriptive**	30	+0.67	A 0.31	+0.21	30 states, below the floor
Gender parity, secondary SE.ENR.SECO.FM.ZS (scored today)	**Descriptive**	22	+0.56	D 0.39	+0.42	22 states, below the floor
Tertiary enrolment SE.TER.ENRR	**Descriptive**	29	+0.60	D 0.80	+0.76	29 states
Adult literacy SE.ADT.LITR.ZS	**Drop**	34	+0.64	G 0.54	+0.53	Near-duplicate of youth literacy
Education spending (% GDP) SE.XPD.TOTL.GD.ZS	**Drop**	46	+0.27	B 0.60	+0.37	An input, not an outcome (Social Progress Index practice); closer to B
Female labour participation SL.TLF.CACT.FE.ZS	**Drop**	53	-0.29	D 0.53	-0.40	Direction contradicted: highest in the poorest agrarian states
Learning poverty SE.LPV.PRIM (scored today)	**Drop**	4	+1.00	A 0.80	+0.40	4 states in 2020-24

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Primary net enrolment SE.PRM.NENR (scored today)	**Drop**	0	—	—	—	No value for any state since 2020; replaced by completion
Secondary net enrolment SE.SEC.NENR (scored today)	**Drop**	0	—	—	—	No value for any state since 2020; replaced by completion
Learning outcomes and out-of-school rates UNESCO UIS (CC BY-SA)	**Ingest**	—	—	—	—	Replaces the lost enrolment series
Social and inter-group trust Afrobarometer	**Context or validation**	—	—	—	—	Horizontal cohesion (Interpeace)

7.4 Pillar D: Health

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Life expectancy SP.DYN.LE00.IN (scored today)	**Keep**	54	+0.76	G 0.65	+0.60	Summary of health; near-duplicate of the mortality rates, so keep one mortality rate beside it
Severe food insecurity SN.ITK.SVFI.ZS (scored today)	**Keep**	43	+0.67	B 0.52	+0.34	Food security (UN component); distinct (margin .15)
Child stunting SH.STA.STNT.ME.ZS	**Add**	54	+0.76	G 0.71	+0.73	Chronic malnutrition; full coverage, development-driven, watch
Malaria incidence SH.MLR.INCD.P3	**Add**	49	+0.64	G 0.60	+0.58	Africa's leading disease burden; margin .04 (Gates-funded MAP refines it)
Maternal mortality SH.STA.MMRT	**Add**	54	+0.73	G 0.69	+0.64	Women's health; less redundant with life expectancy than infant mortality is
Undernourishment SN.ITK.DEFC.ZS	**Add**	50	+0.64	B 0.55	+0.54	Hunger (UN food insecurity); margin .09
Basic sanitation SH.STA.BASS.ZS	**Move to G**	52	+0.72	G 0.75	+0.67	With the other water, sanitation and hygiene measures
Handwashing facilities SH.STA.HYGN.ZS (scored today)	**Move to G**	46	+0.63	G 0.57	+0.77	Water, sanitation and hygiene belong in one pillar: split across D and G they drive D-G (.81)
DPT immunisation SH.IMM.IDPT	**Test**	54	+0.48	G 0.54	+0.33	Near-duplicate of measles: keep at most one
HIV incidence SH.HIV.INCD.ZS	**Test**	49	+0.17	F 0.35	-0.06	A distinct southern-African dimension; weak link to D
Infant mortality SP.DYN.IMRT.IN (scored today)	**Test**	54	+0.79	G 0.66	+0.60	Near-duplicate of under-5 mortality and life expectancy: keep one
Measles immunisation SH.IMM.MEAS	**Test**	54	+0.51	G 0.60	+0.39	Health-system reach; near-duplicate of DPT; closer to G
Moderate or severe food insecurity SN.ITK.MSFI.ZS	**Test**	43	+0.79	G 0.68	+0.52	Overlaps severe food insecurity: keep one
NCD mortality, ages 30-70 SH.DYN.NCOM.ZS	**Test**	54	+0.47	G 0.43	+0.47	Non-communicable disease; margin .04
Nurses and midwives per 1,000 SH.MED.NUMW.P3	**Test**	50	+0.42	G 0.53	+0.60	Health-system capacity; closer to G
Out-of-pocket health spending SH.XPD.OOPC.CH.ZS	**Test**	54	+0.15	A 0.40	+0.15	Financial protection in health; closer to A
Physicians per 1,000 SH.MED.PHYS.ZS	**Test**	51	+0.68	G 0.71	+0.74	Health-system capacity; development-driven, closer to G
Tuberculosis incidence SH.TBS.INCD	**Test**	54	+0.46	F 0.46	+0.16	Disease burden; closer to F
Under-5 mortality SH.DYN.MORT	**Test**	54	+0.81	G 0.75	+0.65	Near-duplicate of infant mortality: keep one

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Safely managed water SH.H2O.SMDW.ZS (scored today)	**Descriptive**	30	+0.88	G 0.82	+0.81	30 states
Air-pollution mortality SH.STA.AIRP.P5	**Drop**	0	—	—	—	No value since 2019
Child wasting SH.STA.WAST.ZS	**Drop**	31	-0.03	C 0.40	+0.03	Direction contradicted and 31 states
Health spending (% GDP) SH.XPD.CHEX.GD.ZS	**Drop**	54	-0.05	C 0.18	-0.08	An input, not an outcome
Disease burden (DALYs) IHME Global Burden of Disease (Gates-funded)	**Non-commercial licence: owner rules**	—	—	—	—	Health outcomes beyond mortality
Household health and nutrition DHS (USAID) and MICS (UNICEF)	**Ingest, terms first**	—	—	—	—	Survey detail, women and children
Malaria prevalence and incidence (modelled) Malaria Atlas Project (Gates-funded)	**Ingest, terms first**	—	—	—	—	Refines the malaria series

7.5 Pillar E: Security and conflict

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Battle deaths (per capita) VC.BTL.DETH	**Replace**	27	+0.54	C 0.34	+0.54	27 states for the same reason; take UCDP directly, where no record means zero
Internally displaced people (per capita) SM.POP.IDPC (scored today)	**Replace**	32	+0.88	A 0.66	+0.38	32 states: WDI omits countries with no displacement, so a gap is often a zero. Use UNHCR/IDMC with explicit zeros
New conflict displacement (per capita) VC.IDP.NWCV	**Replace**	25	+0.82	A 0.63	+0.53	25 states; take IDMC directly with explicit zeros
Homicide rate VC.IHR.PSRC.P5 (scored today)	**Descriptive**	20	+0.20	C 0.35	-0.21	20 states; WDI leaves a gap where UCDP will not
Conflict events ACLED	**Excluded unless terms allow**	—	—	—	—	Restrictive licence
Internal displacement, conflict and disaster IDMC	**Ingest, terms first**	—	—	—	—	Explicit zeros
Military expenditure SIPRI (in WDI)	**Excluded**	—	—	—	—	SIPRI terms differ from CC BY
Non-state conflict deaths (communal violence) UCDP (CC BY 4.0)	**Ingest**	—	—	—	—	Horizontal cohesion
One-sided violence deaths (violence against civilians) UCDP (CC BY 4.0)	**Ingest**	—	—	—	—	Protection of civilians (Interpeace)
Refugees by origin (per capita) UNHCR	**Ingest, terms first**	—	—	—	—	Displacement abroad
State-based conflict deaths UCDP (CC BY 4.0)	**Ingest**	—	—	—	—	Armed conflict, complete with zeros

7.6 Pillar F: Environmental

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Natural resource depletion (% GNI) NY.ADJ.DRES.GN.ZS	**Add**	52	-0.10	A 0.47	+0.23	Depletion (UN definition)
New disaster displacement (per capita) VC.IDP.NWDS	**Add**	49	-0.19	G 0.40	+0.41	Natural disasters (UN definition); per capita
PM2.5 exposure EN.ATM.PM25.MC.M3	**Add**	54	-0.01	C 0.49	+0.17	Degradation (UN definition); full coverage, development-independent
Water stress (SDG 6.4.2) ER.H2O.FWST.ZS	**Add**	53	+0.37	D 0.46	-0.24	Resource depletion (UN definition); SDG 6.4.2
Agricultural land (% land) AG.LND.AGRI.ZS (scored today)	**Test**	54	+0.46	A 0.14	-0.24	Direction undefended: more farmland is not more stable
Cereal yield AG.YLD.CREL.KG	**Test**	52	+0.04	E 0.45	+0.29	Food production capacity; closer to E
Electricity from fossil fuels EG.ELC.FOSL.ZS (scored today)	**Test**	54	+0.75	D 0.40	-0.33	Distinct (margin .35) but energy mix is not environmental insecurity
Forest area (% land) AG.LND.FRST.ZS	**Test**	54	+0.14	D 0.26	-0.04	Use the change in forest area (degradation), not its level
Freshwater withdrawal (% resources) ER.H2O.FWTL.ZS (scored today)	**Test**	53	+0.53	D 0.53	-0.28	Near-duplicate of water stress: keep the SDG measure
Protected land ER.LND.PTLD.ZS	**Test**	54	+0.16	E 0.33	0.00	Protection effort, not insecurity
CO2 per capita EN.GHG.CO2.PC.CE.AR5 (scored today)	**Descriptive**	53	+0.63	D 0.77	-0.93	A wealth proxy (GDP rho -.93); approved plan 4.2
Renewable energy share EG.FEC.RNEW.ZS	**Drop**	54	+0.53	D 0.67	-0.70	An inverse wealth proxy like CO2 (GDP rho -.70)
Climate exposure and sensitivity ND-GAIN	**Ingest, terms first**	—	—	—	—	Climate pressure (Interpeace)
Disaster events and people affected EM-DAT, UCLouvain	**Ingest, terms first (registration)**	—	—	—	—	Natural disasters (UN definition)
Natural hazard exposure INFORM Risk (EC JRC, UN OCHA)	**Ingest, terms first**	—	—	—	—	Disasters

7.7 Pillar G: Structural and infrastructure

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Basic drinking water SH.H2O.BASW.ZS (scored today)	**Keep**	52	+0.80	D 0.73	+0.77	Core infrastructure; joined by sanitation and handwashing from D
Electricity access EG.ELC.ACCS.ZS (scored today)	**Keep**	54	+0.83	D 0.77	+0.81	Core infrastructure; development-driven
Internet users IT.NET.USER.ZS	**Add**	52	+0.81	D 0.77	+0.87	Connectivity of information (Interpeace); development-driven, watch
Mobile subscriptions IT.CEL.SETS.P2	**Add**	54	+0.65	D 0.61	+0.56	Connectivity; full coverage, less development-driven (.56)
Bank branches FB.CBK.BRCH.P5	**Test**	48	+0.71	D 0.74	+0.71	Closer to D; development-driven
Clean cooking fuels EG.CFT.ACCS.ZS	**Test**	54	+0.71	D 0.72	+0.84	Development-driven; closer to D

Indicator	Status	n	Own ρ	Closest other	GDP ρ	Why
Fixed broadband IT.NET.BBND.P2	**Test**	54	+0.82	D 0.81	+0.75	Near-duplicate of internet users
Power transmission losses EG.ELC.LOSS.ZS	**Test**	37	0.00	B 0.14	+0.06	Quality of the grid, not just access; correlated with nothing
Logistics performance LP.LPI.OVRL.XQ	**Descriptive**	30	+0.42	B 0.50	+0.30	30 states
Safety-net coverage per_sa_allsa.cov_pop_tot	**Descriptive**	13	+0.15	F 0.40	-0.18	13 states
Social insurance coverage per_si_allsi.cov_pop_tot	**Descriptive**	12	+0.59	D 0.59	+0.67	12 states
Social protection coverage per_allsp.cov_pop_tot (scored today)	**Descriptive**	13	+0.15	D 0.36	-0.11	13 states in 2020-24

The full list, with every correlation, is also in *ASI indicator candidates.csv* beside this document.

8. After the pillars are built: the statistics we run

Stage	Method	What it gives you
Normalise	Min-max to 0–100 against goalposts set from observed values; outliers capped; logs where the skew test says so	Indicators on one scale, stable over time
Aggregate	Arithmetic mean within a pillar; equal weights across pillars; geometric mean as a check	The pillar and composite scores
Check weights	Effective weight of each indicator, after correlations	Proof that equal means equal
Reliability	Coverage tiers; an INFORM-style reliability index	How much of each score is measured
Intervals	Each composite as the mean of its pillars: SE = SD/√k, t interval (df = k − 1)	A confidence interval per country
Rank bands	z test on each pair's difference (P46, P48)	Tiers rather than 54 point ranks
Sensitivity	Monte Carlo over weights, missing-data rules and goalposts (JRC practice)	How far ranks move under other reasonable choices
Development view	Pillars with log GDP per capita held constant	Stability beyond income
External validation	Spearman ρ with IIAG, FSI, GPI and WGI political stability, each with its Fisher interval and n	Whether others see the same picture
Edition monitoring	Two-correlation Z test between editions; the rebuild check	Whether a change is real or noise

9. Implementation

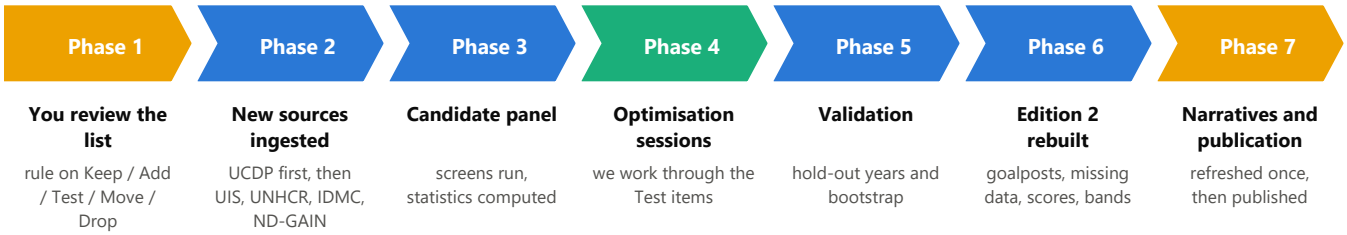


Figure 6. The phases. Gold: you decide. Green: we work together. Blue: the agent builds.

Phase	What happens	Your part
1 Review	You go through section 7 and the CSV	Rule on each status; add indicators you want
2 Ingest	UCDP first (CC BY 4.0), then UIS, UNHCR, IDMC, ND-GAIN, EM-DAT; terms read and dated before use	Rule on IHME's non-commercial licence
3 Panel	All approved candidates in one panel; screens run	—
4 Optimise	Sessions on the Test items: each proposed move with its statistics	Approve or reject each move
5 Validate	Other years and bootstrap	Sign off the frozen structure
6 Rebuild	Edition 2 on the new structure: goalposts, missing-data rule, scores, intervals, bands, divergence report	Review the report
7 Publish	Narratives refreshed once; site rebuilt	Approve publication

An idea for the sessions: a small interactive page where each candidate can be switched in or out and moved between pillars, with the correlation matrix, HTMT and coverage updating live, so you can see each trade-off before approving it.

This redesign absorbs three steps of the approved plan: the admission floor (4.12), the rebuild of Pillar F (4.2) and the end of WGI cross-listing (4.3). The missing-data rule, goalposts and bands follow in phase 6.

10. The sources, evaluated

Source	Adds	Licence	Verdict
World Bank WDI	1,498 series; most candidates	CC BY 4.0 (read 2026-10-17)	In Athena
World Bank WGI	Six governance estimates with standard errors	CC BY 4.0	In Athena
UCDP (Uppsala)	State-based, non-state and one-sided violence deaths, with zeros	CC BY 4.0 per its download centre	Ingest first
UNESCO UIS	Learning outcomes, out-of-school rates	CC BY-SA 4.0 (read 2026-09-26)	Ingest
UNHCR	Refugees by origin	To read	Ingest, terms first
IDMC	Displacement by conflict and disaster, with zeros	To read	Ingest, terms first
ND-GAIN	Climate exposure and capacity	To read	Ingest, terms first
EM-DAT	Disasters and people affected	Registration; to read	Ingest, terms first
INFORM Risk	Hazard exposure	To read	Ingest, terms first
V-Dem	Participation, egalitarian access to power	To read	Ingest, terms first
Ethnic Power Relations	Groups excluded from power	To read	Ingest, terms first
World Justice Project	Rule of law as experienced	To read	Ingest, terms first
Afrobarometer	Trust in institutions and between groups	Free, copyrighted, citation and copies of publications	Context or validation (about 39 states, every 2–3 years)
Malaria Atlas Project (Gates-funded)	Modelled malaria burden	To read	Ingest, terms first
IHME Global Burden of Disease (Gates-funded)	Disease burden beyond mortality	Free, non-commercial only	Your ruling
DHS and MICS	Household health, women and children	Registration; to read	Ingest, terms first
Gates Foundation	Open-access data from research it funds	Usually CC BY, dataset by dataset	Through the named datasets
IMF WEO	Government debt and revenue	To read	Ingest, terms first
FAOSTAT	Food prices and production	To read	Ingest, terms first
IIAG, FSI, GPI	Other indices	To read	Validation only
ACLED	Conflict events	Corporate licence for commercial use; no redistribution	Excluded unless terms allow
SIPRI military spending	In WDI under SIPRI's own terms	Not CC BY	Excluded

11. Decisions for you

- **The target (section 2.2).** Discriminant validity with HTMT below .85 and every indicator closer to its own pillar, rather than zero correlation. Agree, or set a different bar.
- **The rules (section 6.1).** At least 4 indicators and 2 source families per pillar; the 33-state floor; acceptance only if a change holds in other years.
- **The list (section 7).** Each status, especially the Test items and the moves of water, sanitation and financial inclusion into G.
- **Total GDP.** Scored in Pillar B on a log scale, as you ruled; confirm current US dollars.
- **The development-adjusted view.** Publish it beside the main index, or keep it internal.
- **IHME's non-commercial licence.** Acceptable for this index or not.

Glossary

Term	Meaning
Pillar	One of the seven dimensions; the mean of its indicators
Cross-listing	One indicator counted in two pillars
Convergent validity	An indicator agrees with the rest of its own pillar
Discriminant validity	A pillar measures something the others do not
HTMT	Heterotrait-monotrait ratio: correlation between pillars relative to within them
Third variable	A variable that drives two others, making them look related (here, development)
Partial correlation	The correlation left after removing a third variable from both
Attenuation	Measurement error shrinking a correlation; correcting it estimates the true one
Second-order factor	A factor (stability) behind several factors (the pillars)
Bootstrap	Re-computing a statistic on many resamples of the countries to see how much it varies
Descriptive	Shown in the dashboard, not counted in the score